

Physics Lecture 1: Classical Mechanics

Introduction to Newtonian Mechanics

Classical mechanics is the branch of physics that describes the motion of macroscopic objects, from projectiles to astronomical objects, under the influence of forces. It is one of the oldest and largest subjects in science, and it serves as the foundation for much of modern technology and engineering.

Key Concepts and Formulas

- **Newton's First Law (Law of Inertia):** An object at rest stays at rest and an object in motion stays in motion with the same speed and in the same direction unless acted upon by an unbalanced force.
- **Newton's Second Law (The Law of Acceleration):** The acceleration of an object as produced by a net force is directly proportional to the magnitude of the net force, in the same direction as the net force, and inversely proportional to the mass of the object. $\Sigma F = ma$
Here, F is the net force, m is the mass, and a is the acceleration. This is a vector equation, meaning both magnitude and direction must be considered.
- **Newton's Third Law (Law of Action and Reaction):** For every action, there is an equal and opposite reaction. If object A exerts a force on object B, then object B must exert a force of equal magnitude and opposite direction back on object A. $F_{AB} = -F_{BA}$
- **Kinematic Equations:** These equations describe motion with constant acceleration.
 $v = v_0 + at$
 $x = x_0 + v_0t + \frac{1}{2}at^2$
 $v^2 = v_0^2 + 2a(x - x_0)$
 - v = final velocity
 - v_0 = initial velocity
 - a = constant acceleration
 - t = time
 - x = final position
 - x_0 = initial position

Mathematical Example Problem: Projectile Motion

Problem: A ball is thrown from the ground with an initial velocity of 20 m/s at an angle of 30° above the horizontal. Assuming negligible air resistance, what is the maximum height the ball reaches?

Solution:

1. **Deconstruct the initial velocity:** The initial velocity vector v_0 has both a horizontal (v_{0x}) and vertical (v_{0y}) component.
 $v_{0x} = v_0 \cos(\theta) = 20 \cos(30^\circ) = 17.32 \text{ m/s}$
 $v_{0y} = v_0 \sin(\theta) = 20 \sin(30^\circ) = 10 \text{ m/s}$
2. **Analyze vertical motion:** At the maximum height, the vertical velocity of the ball is

0 m/s. We can use the kinematic equation $v^2 = v_0^2 + 2a(x - x_0)$ to find the maximum height, $y_{\max}$. We will use the vertical components for this analysis. $v_y^2 = v_{0y}^2 + 2a_y(y_{\max} - y_0)$

We know:

- $v_y = 0$ m/s (at the peak)
- $v_{0y} = 10$ m/s
- $a_y = -9.8$ m/s² (acceleration due to gravity, negative because it acts downwards)
- $y_0 = 0$ m (initial height)

3. **Solve for**

$$y_{\max}: 0^2 = (10)^2 + 2(-9.8)(y_{\max} - 0) \quad 0 = 100 - 19.6y_{\max} \quad 19.6y_{\max} = 100 \quad y_{\max} = \frac{100}{19.6} \approx 5.1 \text{ m}$$

The maximum height the ball reaches is approximately 5.1 meters.

This concludes our first lecture on Classical Mechanics.